

# From Classical Encryption to Post-Quantum Security: A Study of Lattice-Based Cryptography

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## INTRODUCTION

Cryptography protects digital communication and has evolved from simple methods like the Caesar cipher to advanced mathematical systems.

Modern systems such as RSA rely on hard problems like factorization, but quantum computing threatens them through algorithms like Shor's algorithm.

To address this, post-quantum cryptography has been developed. A leading approach is lattice-based cryptography, which uses hard problems like SVP and LWE and is believed to remain secure against quantum attacks.

## METHODS

This study uses both theoretical mathematical modeling and computational simulations to analyze cryptographic systems. Classical encryption is demonstrated using the Caesar cipher, which applies modular arithmetic to shift letters in the alphabet.

Lattice-based cryptography is modeled by constructing lattices from integer combinations of basis vectors, which are also visualized in two dimensions to show their geometric structure.

The Learning With Errors (LWE) encryption system is defined as  $b = As + e \pmod{q}$ , where  $A$  is a public matrix,  $s$  is the secret key,  $e$  is a noise vector, and  $q$  is the modulus.

To test security, an attack simulation is performed by randomly guessing the secret key and measuring the error using

$$\|As_{guess} - b\|$$

This helps evaluate how resistant the system is to brute-force attacks.

## DATA

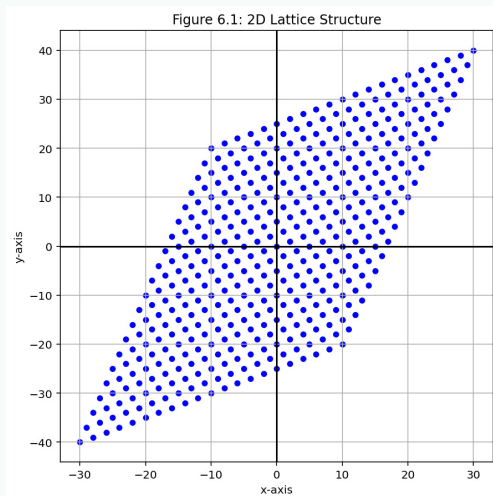


Figure 1

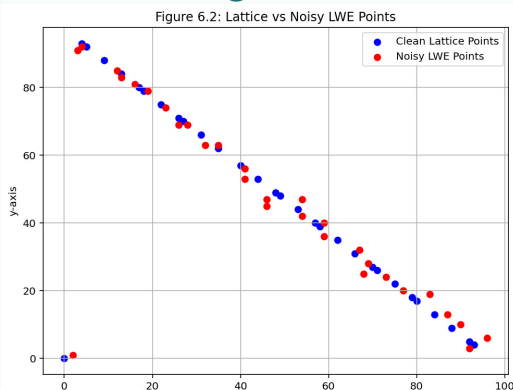


Figure 2

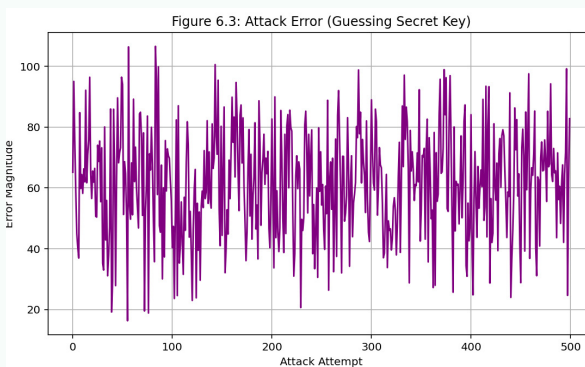


Figure 3

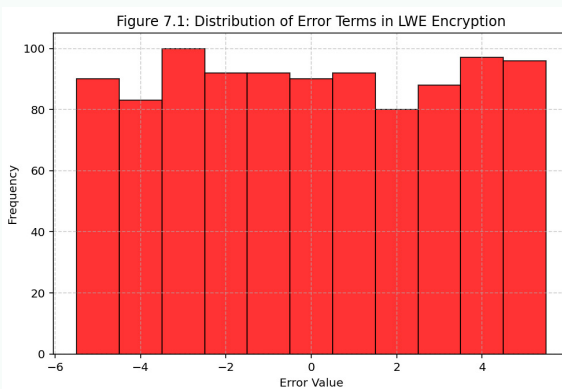


Figure 4

Figure 1: Lattice Structure  
Shows a repeating grid of points generated from basis vectors  
Represents structured but high-dimensional geometry in cryptography. In higher dimensions, this structure becomes extremely complex and hard to solve\

Figure 2: Lattice vs Noisy LWE Points  
Two sets of points:

Clean lattice points (structured pattern). Noisy LWE points (randomly displaced). Key Insight: Noise destroys structure, making reverse-engineering impossible

Figure 3: Attack Error Graph  
Shows error values from random guesses of the secret key  
Error fluctuates with no clear pattern  
No convergence to the correct key (attack fails)

Figure 4: Noise Distribution (LWE Error Term)  
Shows distribution of error vector  $e$  in LWE. Noise is random but bounded. This randomness makes the system secure against attacks

## ANALYSIS

The results highlight clear differences between classical and lattice-based cryptography.

Classical systems (like the Caesar cipher) are simple and easily broken. In contrast, lattice-based cryptography relies on high-dimensional geometry and noise, making it computationally difficult to solve.

The LWE model adds controlled randomness, which hides the secret key and prevents reconstruction. Simulations show that random guessing fails, error values remain unstable, and brute-force attacks are ineffective.

Overall, lattice-based problems lack the structure needed for quantum algorithms like Shor's algorithm to exploit, making them resistant to both classical and quantum attacks.

## CONCLUSION

This study shows that classical cryptographic systems are vulnerable to quantum computing, while lattice-based cryptography offers a strong alternative for the post-quantum era. By relying on mathematical structures such as lattices and hard problems like LWE, it achieves security based on geometric complexity rather than simple arithmetic difficulty. Simulation results further confirm that lattice-based systems resist both classical and quantum attacks, making them a leading candidate for future secure communication.